Bull. Mater. Sci. (2026) 49:113
https://doi.org/10.1007/s12034-026-03623-6

Fabrication of lunar basalt bulk using the DC FHPs method

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MS received 29 September 2025; accepted 5 December 2025

Abstract. With the advancement of lunar exploration programs, lunar basalt materials have demonstrated significant potential for aerospace, construction, and other applications due to their high strength, thermal stability and other exceptional properties. However, their unique characteristics and the constraints of extraterrestrial environments pose challenges for efficient fabrication. This study successfully achieved the rapid preparation of bulk lunar basalt materials using the direct current (DC) fast hot-pressed sintering (FHPs) method. The phase composition, microstructure and mechanical properties of the sintered samples were systematically analyzed through X-ray diffraction (XRD), scanning electron microscopy (SEM), density measurements, Vickers hardness testing and three-point bending tests. The results indicated that the optimal material performance was achieved under sintering conditions of 1000°C (maximum temperature), 40 MPa (applied pressure), and a 10-min holding time. The resulting samples exhibited a density of 3.12 g cm^{-3}, a Vickers hardness of 853.7 HV, a fracture toughness of 2.2 MPa $m^{1/2}$ and bending strength of 85 MPa, demonstrating excellent microstructural integrity and mechanical properties. Another large-sized sample of 210 mm × 110 mm × 10 mm was prepared at 960°C under 30 MPa for 20 min, which provided both theoretical understanding and engineering validation support for the *in-situ* utilization of lunar basalt in lunar base construction.

Keywords. Lunar basalt; direct current fast hot-pressed sintering; microstructure; mechanical properties.

1. Introduction

As a core goal of deep space exploration for various countries, lunar base construction faces a key bottleneck in the efficient utilization of *in-situ* resources. The cost of Earth–Moon transportation is as high as \$50,000–\$90,000 per kilogram, making the *in-situ* conversion of lunar soil into construction materials an inevitable choice [1,2]. Basalt, the main component of lunar soil (accounting for more than 60% of the material in lunar mare regions), has a mineral composition similar to that of terrestrial basalt. It mainly consists of plagioclase ($NaAlSi_3O_8$–$CaAl_2Si_2O_8$), pyroxene ($(Ca, Na)(Mg, Fe, Al)(Si, Al)_2O_6$), magnesium-rich olivine ($(Mg, Fe)_2SiO_4$) and small amounts of minerals such as ilmenite ($FeTiO_3$) and magnetite (Fe_3O_4). Thus, it is regarded as a core raw material for preparing structural materials [3–6]. Through simulated lunar soil sintering experiments, the 'Lunar Soil Brick' project of the China National Space Administration determined that 1000°–1100°C is the optimal sintering temperature, and the obtained specimens have a compressive strength of 20 MPa, meeting the initial requirements for basic building materials of lunar bases [7]. Zhou *et al* [8] revealed the melting behavior of lunar soil particles at 1000°–1100°C through vacuum hot-pressed sintering experiments, and found that the co-melting effect of ilmenite and plagioclase could increase the density of sintered bulk materials to over 92%. Suhaizan *et al* [9] proposed the 'Yuehu Zun' architectural scheme, innovatively combining basalt fiber reinforcement technology with 3D printing technology, and realizing modular construction through mortise and tenon structure design, which provided a new idea for *in-situ* construction in a microgravity environment.

Based on in-depth analysis of compositional characteristics, the engineering preparation of lunar soil basalt has become a core link in resource utilization. Currently, the preparation technologies of lunar soil basalt materials are mainly divided into two categories: chemical solidification and physical sintering. In terms of chemical solidification, basalt simulated lunar soil and slag are mainly used as raw materials, and solidification is achieved through alkali activation technology, with a compressive strength of up to 61.3 MPa. However, this technology relies on the introduction of alkali activators, which limits the utilization rate

Published online: 15 June 2026

of *in-situ* resources [10]. In terms of physical sintering, the existing technologies, such as laser sintering and microwave sintering, can realize the melting and bonding of lunar soil particles, but they have problems such as high energy consumption and low efficiency, which make it difficult to meet the needs of low-energy consumption and rapid construction on the Moon [11,12]. In addition, the lunar surface environment, with a vacuum ($\leq 10^{-14}$ Pa) and extreme temperature differences (−250° to 127°C), will significantly affect the material sintering process. Hu *et al* [13] found that the viscosity and structural stability of simulated lunar soil glass melt under vacuum conditions were significantly different from those under normal pressure, and the ground test results of traditional sintering processes could not be directly transferred to the lunar surface. Although additive manufacturing technology provides a new path for *in-situ* forming of lunar soil, such as the micro-extrusion 3D printing proposed by Wu *et al* [14], the fluidity and solidification strength of the 'ink' of printing materials still depend on particle gradation optimization, and insufficient interlayer bonding strength remains a bottleneck. In contrast, the hot-pressed sintering method can quickly achieve particle densification through the synergistic effect of pressure and temperature, but its traditional process has problems, including low heating rate (usually <10°C min^{-1}) and large equipment volume, which cannot adapt to the needs of miniaturized construction equipment on the Moon [15]. The direct current (DC) fast hot-pressed sintering (FHPs) method [16], relying on the 'DC Joule Heating' characteristic, can achieve a fast heating rate of 1000°C min^{-1}-surpassing spark plasma sintering (SPS, 500°–800°C min^{-1}). In terms of scalability, FHPs can fabricate large-sized square samples (e.g., 210 mm × 110 mm × 10 mm in this study) without obvious performance degradation, whereas SPS is typically limited to small cylindrical samples (<100 mm diameter). For energy efficiency, FHPs directly use DC power from lunar solar systems with 85% energy conversion efficiency, compared to 60–70% for SPS (due to pulse-current energy loss) and <40% for laser sintering (due to laser reflection and scattering). These advantages make FHPs more suitable for extraterrestrial *in-situ* resource utilization (ISRU) than conventional field-assisted sintering technologies. Applying FHPs to the preparation of lunar soil basalt materials is expected to address the problems of high energy consumption and low efficiency associated with traditional technologies. On the one hand, DC heating can directly use the DC from the lunar solar power generation system, reducing energy conversion loss. On the other hand, rapid sintering can reduce volatile loss in the lunar surface vacuum environment.

Based on the above researches, this study uses FHPs method to prepare lunar soil basalt bulk materials, solves the key problems of process parameter optimization, performance improvement and large-size sample preparation in the preparation of lunar soil basalt bulk materials, and provides a practical technical path for the efficient utilization of lunar soil basalt. It focuses on exploring the effects of sintering temperature (900°–1000°C), pressure (30–40 MPa) and holding time (10–20 min) on the material phase composition, microstructure (such as porosity, crystal bonding state) and mechanical properties (density, hardness, flexural strength, etc.). A suitable process parameter system is established to solve the problems of low efficiency and poor adaptability of traditional technologies, providing technical support for the *in-situ* preparation of high-performance lunar soil basalt bulk materials.

2. Experimental

The lunar basalt powder used in this experiment was derived from simulated materials based on the samples collected by the Chang'e series probes. Its primary chemical composition closely resembles that of authentic lunar basalt, ensuring the experimental representativeness. In this study, the samples were fabricated using a DC FHPs (FHPs-828/858, Hateng) method (see process curve in figure 1a, b). The lunar basalt powder was loaded into a Φ20 mm graphite mold and sintered under the following conditions. The heating rate was 100°C min^{-1}; vacuum level was <10 Pa; sintering temperature was 900°–1000°C; pressure was 40 MPa and holding time was 10 min. For larger-scale samples, the powder was loaded into 80 mm × 55 mm ×10 mm and 210 mm × 110 mm ×10 mm square graphite molds and processed using FHPs-858 equipment. The parameters were adjusted as follows. Heating rate was 100°C min^{-1}; vacuum level was <10 Pa; sintering temperature was 960°–1000°C; pressure was 30–40 MPa; and holding time was 10–20 min.

X-ray diffraction (XRD) analysis was performed using a D/Max-2500/pc diffractometer (Rigaku) with Cu Kα radiation (λ = 0.15406 nm). Measurements were conducted in step-scan mode over a 2θ range of 10°–80° with a step size of 0.02°. Fracture morphology was examined using a TESCAN Mira field-emission scanning electron microscope (SEM) operated at an acceleration voltage of 20 kV. The sample density was determined by the Archimedes method using distilled water as the immersion medium, achieving a measurement accuracy of ±0.01 g cm^{-3}. The Vickers hardness was measured using an HMV-G21ST microhardness tester. The 10 valid indentations were made on each sample, and the results were averaged. The three-point bending tests were performed using an HF-9006 universal testing machine with a 30 mm support span and 0.5 mm min^{-1} loading rate. The 80 mm × 55 mm × 10 mm samples were sectioned into 10 test bars (35 mm × 3 mm × 4 mm). All surfaces were polished, and the four long edges were chamfered at 45° (0.1 mm depth). Additional test bars (35 mm × 3 mm × 4 mm) were prepared as above. A penetrating pre-crack (1.5 mm depth, ≤0.2 mm width) was introduced at the specimen mid-span using wire EDM to simulate actual fracture conditions. Fracture toughness

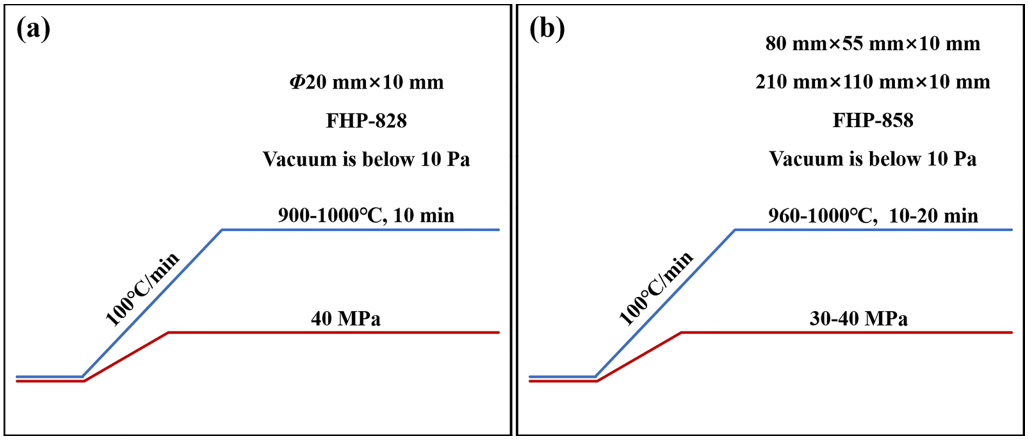

Figure 1. The temperature-pressure-time profile of the FHPs process: (**a**) Sintering process curve for small-sized samples; (**b**) Sintering process curve for large-sized samples.

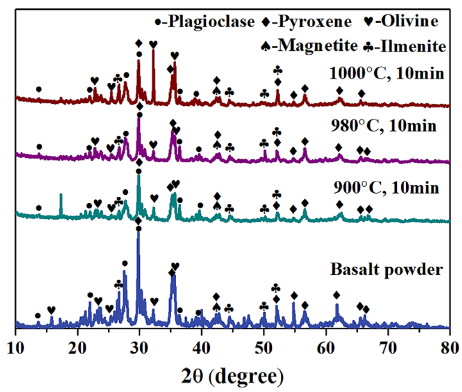

Figure 2. XRD spectra of lunar basalt powder and bulk material.

(K_{IC}) was determined from the maximum load (P) at failure using standard fracture mechanics formulae, incorporating the pre-crack length (a), critical load (P), span (S), specimen thickness (B), width (W) and shape factor $\left(f\left(\frac{a}{W}\right)\right)$. The testing parameters matched those of the flexural strength tests. The formula was

$$K_{IC} = \frac{P \cdot S}{B \cdot W^{3/2}} \cdot f\left(\frac{a}{W}\right).$$

3. Results and discussion

Phase analysis of the lunar basalt powder (figure 2) reveals that its primary mineral composition consists of plagioclase (PDF#41-1486), pyroxene (PDF#24-0202), and olivine (PDF#34-0179), with minor phases including magnetite (PDF#19-0629), ilmenite (PDF#29-0733) and other accessory minerals. To further investigate the microstructural characteristics of the powder, SEM images and EDS surface scanning were employed. The SEM micrograph (figure 3a) demonstrates that the particle surfaces exhibit irregular undulations and striated textures, reflecting the inherent heterogeneity of this multi-mineral system inherited from the parent rock.

SEM analysis reveals a clear sintering evolution pathway. At 900°C (figure 3b), the particles remain in a loosely packed state with large-sized pores unevenly distributed, resulting in a low sintering density. As the temperature increases to 980°C (figure 3c), interparticle necking becomes significantly more pronounced, accompanied by a simultaneous reduction in both the number and size of pores, thereby accelerating the densification process. Upon reaching 1000°C (figure 3d), a continuous molten layered structure forms with pores nearly eliminated and particles strongly bonded via interfacial diffusion, indicating a substantial improvement in sintering density.

EDS surface scanning analysis of the lunar basalt bulk sintered at 1000°C (figure 4) demonstrates the intrinsic relationship between elemental migration, quantitative grain size evolution, and mineral phase evolution. The continuous and homogeneous distribution of Si confirms the establishment of a stable silicate network framework at high temperatures, primarily composed of feldspar and pyroxene. The strong spatial correlation among Ca, Al, and Si distributions not only aligns with the lattice structure of plagioclase (average grain size 5–8 μm, with long axes of 8–12 μm and short axes of 3–5 μm), but also corroborates the enhanced crystallinity observed in XRD patterns. The enrichment zones of Fe and Mg coincide with the molten layered structure observed in SEM, indicating that ferromagnesian minerals, pyroxene with an average grain size of 3–5 μm, and olivine (as the Fe–Mg strong enrichment phase) with a slightly smaller grain size of 2–4 μm, undergo melting and recombination, thereby enhancing intergranular bonding. Ti exhibits a discrete punctate distribution, reflecting the incomplete melting of ilmenite and other accessory minerals during sintering. The migration and recombination of major elements (Si, Al, Ca, Mg, Fe) govern particle interfacial fusion, leading to the formation of a dense silicate network with coordinated grain growth

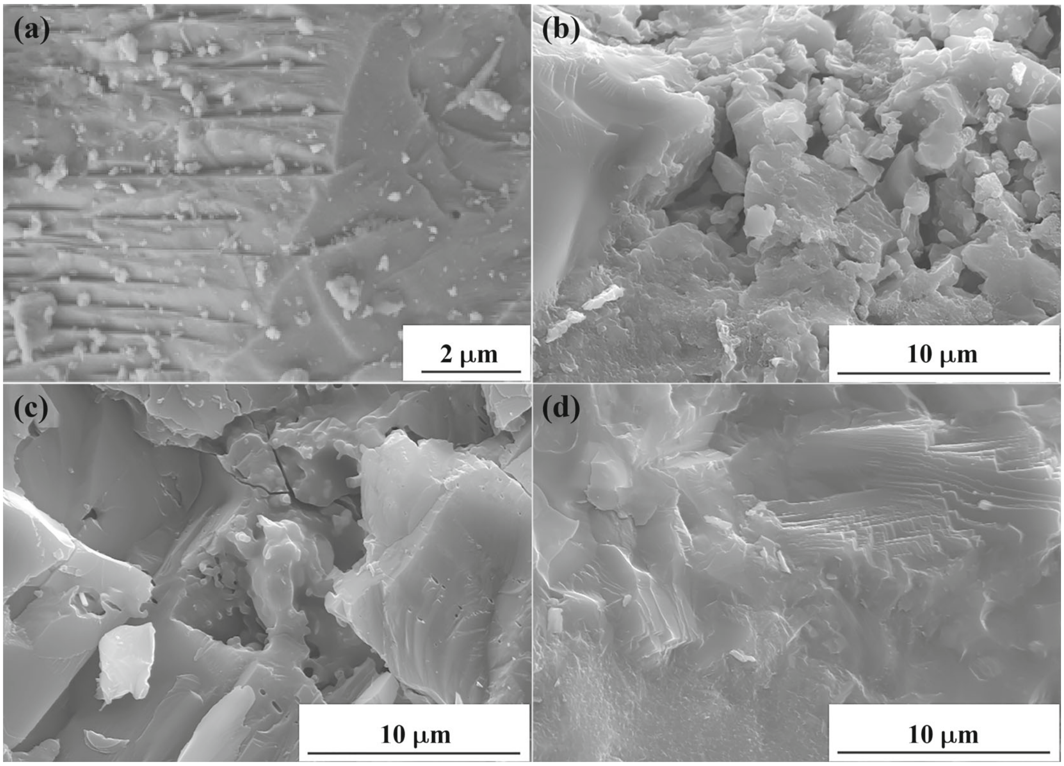

Figure 3. Basalt powder and cross-sectional SEM micrographs of the as-synthesized basalt blocks under varying sintering conditions (constant 40 MPa pressure and 10 min duration): (**a**) Basalt powder; (**b**) 900°C; (**c**) 980°C; (**d**) 1000°C.

across mineral phases ranging from 2–8 μm. The localized trace element enrichments (Na, Ti) preserve evolutionary signatures of minor mineral phases, while the distinct grain size gradients among plagioclase, pyroxene and olivine further illustrate the differential sintering behavior and phase evolution mechanisms under high-temperature conditions.

Multi-scale characterization (XRD, SEM, EDS) clarifies the high-temperature densification mechanism via melting, diffusion and recrystallization. The chemical element composition of the sintered simulated basalt block was determined by EDS, and the results are shown in table 1. A comparison of the main chemical components between the simulated lunar basalt and the real lunar basalt reveals that the former has a higher silicon content and a lower iron content. Despite this numerical difference, the basalt raw material used contains the core elements of lunar soil, which verifies the effectiveness of the simulated lunar soil to a certain extent. Analysis indicates that the chemical composition of basalt varies significantly across different production areas. Moreover, due to differences in their geological evolution processes, the chemical compositions of basalt on Earth and the Moon also show obvious distinctions. Therefore, screening raw materials whose mineral composition and chemical composition are both closest to those of the Moon is an engineering task with relatively great difficulty.

Figure 5a–d presents the performance measurement of basalt bulk samples synthesized under constant pressure (40 MPa) with 10 min heat preservation. The results demonstrate a significant temperature dependence of the material properties. As the sintering temperature increased from 900° to 1000°C, the bulk density exhibited a progressive enhancement from 2.75 to 3.12 g cm^{-3} (figure 5a), achieving 95% of the theoretical density (3.3 g cm^{-3}) [9]. This density evolution suggests progressive particle fusion and densification with concomitant porosity reduction at elevated temperatures. Concurrently, the Vickers hardness showed a 6.3% improvement from 803.4 to 853.7 HV (figure 5b). The observed hardness enhancement correlates with the increased structural densification, indicating stronger inter-crystalline bonding and improved resistance to mechanical indentation at higher sintering temperatures.

The sample sintered at 980°C exhibits a density of 3.01 g cm^{-3} and a Vickers hardness of 837.9 HV, closely approaching the values obtained at 1000°C (3.12 g cm^{-3} and 853.7 HV). This suggests that near-complete densification is achieved at 980°C, with only marginal improvements observed at higher temperatures. At this stage, the internal pore structure is substantially reduced, particle bonding reaches optimal compactness, and the material's mechanical properties converge with those of samples sintered at elevated temperatures.

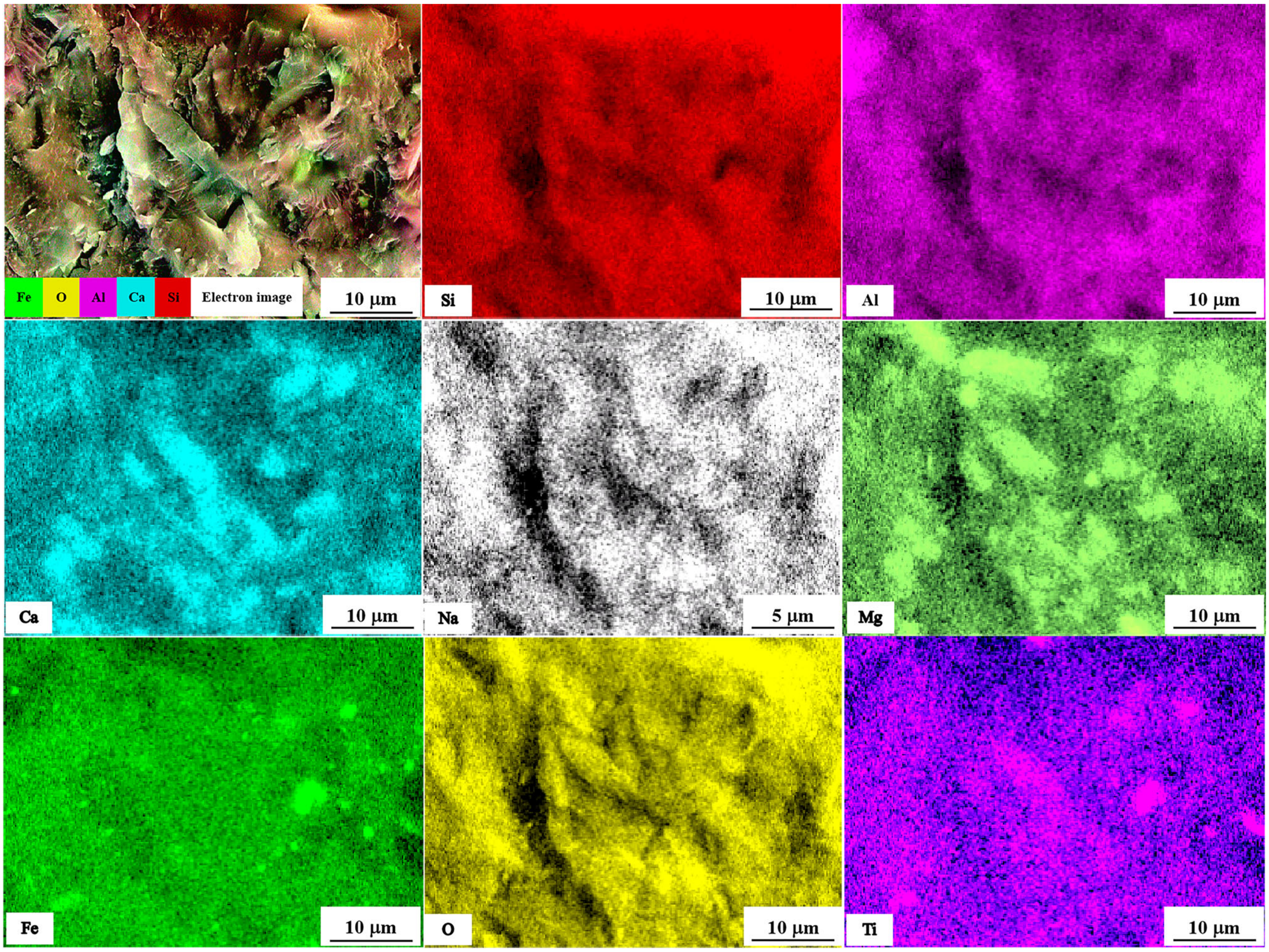

Figure 4. EDS elemental mapping of the cross-section of the basalt synthesized at 1000°C under 40 MPa pressure for 10 min.

Table 1. Chemical composition of lunar basalt [17].

Site	SiO_2 (wt%)	Al_2O_3 (wt%)	CaO (wt%)	FeO (wt%)	MgO (wt%)	TiO_2 (wt%)
Apollo11	40.0	11.6	10.9	18.7	9.6	8.3
Apollo12	45.2	14.9	11.5	15.5	9.3	2.6
Apollo14	45.4	17.1	11.7	11.5	10.1	2.4
Apollo16	45.1	27.3	15.7	5.3	5.9	0.7
Apollo17	39.9	11.0	10.5	17.7	9.8	9.39
Chang'E-3	42.8	11.9	10.6	21.6	9.6	4.1
Chang'E-5	42.2	11.60	10.90	22.20	5.80	5.70
Simulated lunar basalt	58.3	18.5	11.5	14.7	4.0	4.3

The mechanical properties of the sintered basalt samples (80 mm × 55 mm × 10 mm) were further evaluated under optimized conditions (980°–1000°C, 40 MPa, 10 min), demonstrating a bending strength of up to 85 MPa (figure 5c). Fracture toughness analysis revealed a systematic increase in K_{IC} with sintering temperature. Specimens sintered at 980°C exhibited a K_{IC} of 2.1 MPa $m^{1/2}$, while those sintered at 1000°C reached 2.2 MPa $m^{1/2}$ (figure 5d). This trend aligns with the observed enhancements in density and hardness, suggesting a unified densification mechanism. Progressive densification not only mitigates initial defects (e.g., pores and microcracks) that serve as crack initiation sites but also enhances intergranular bonding, thereby restricting unstable crack propagation. Cross-referencing with density and hardness data confirms that the material achieves optimal microstructural synergy at 1000°C. At this

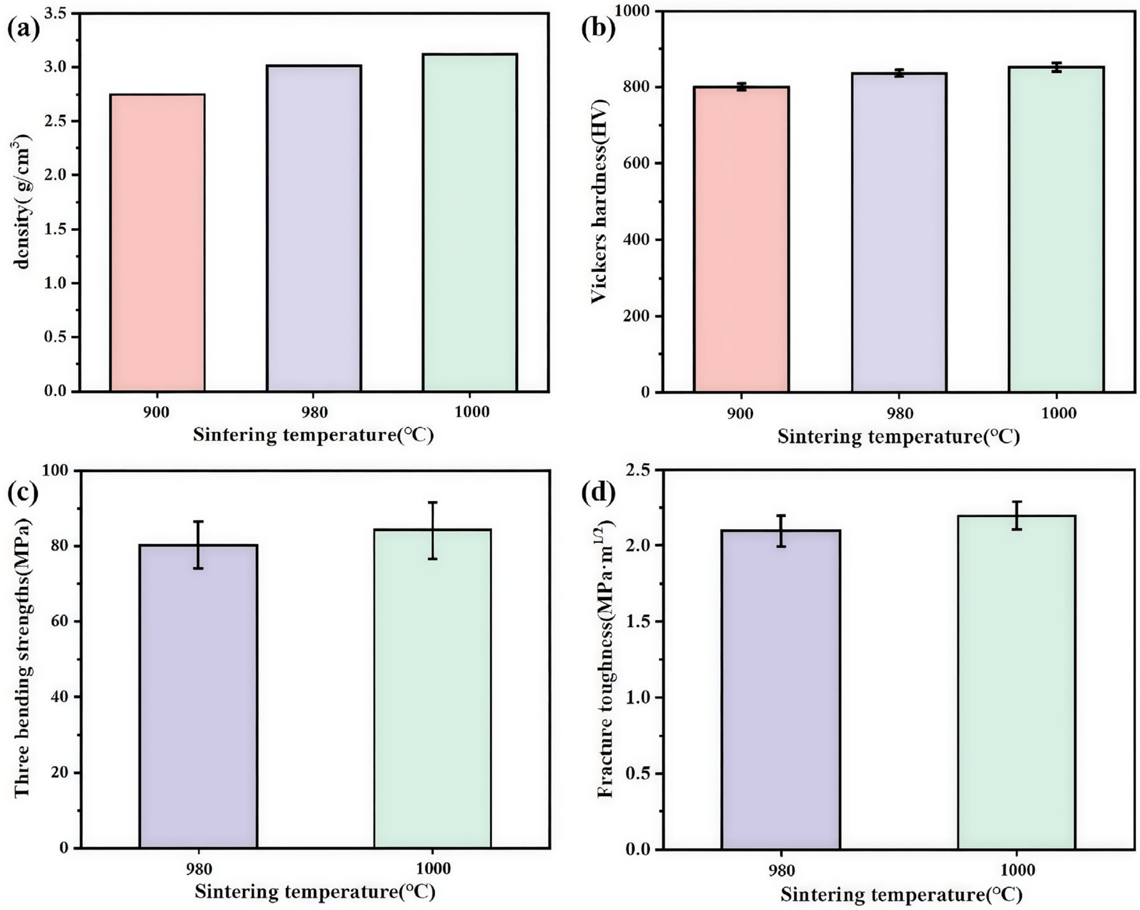

Figure 5. Performance measurement of synthesized basalt bulk samples at varying sintering temperatures (40 MPa holding pressure, 10 min heat preservation): (**a**) Density; (**b**) Vickers hardness; (**c**) Three bending strengths; (**d**) Fracture toughness.

stage, the moderate grain coarsening and high relative density (95%) collectively enhance mechanical performance. Grain coarsening reduces grain boundary density, minimizing stress concentration points. Near-full densification eliminates internal defects, promoting homogeneous stress distribution under load. This synergistic effect results in improved bending strength and fracture resistance, as stress is more effectively dissipated throughout the bulk material.

The FHPs method achieves rapid material densification through the synergistic coupling of a DC electric field and dynamic pressure. This process is governed by three key mechanisms. (1) Joule heating-dominated rapid sintering: When a DC current passes through the graphite die and the powder compact, resistive Joule heating occurs at a rate exceeding 1000°C min^{-1}. This ultra-fast heating significantly reduces atomic diffusion activation times, promoting eutectic reactions among lunar basalt constituents (e.g., plagioclase–pyroxene–ilmenite systems). At 1000°C, the generated eutectic liquid phase rapidly infiltrates particle boundaries, dramatically accelerating densification kinetics [8]. (2) Pressure-controlled microstructure optimization:

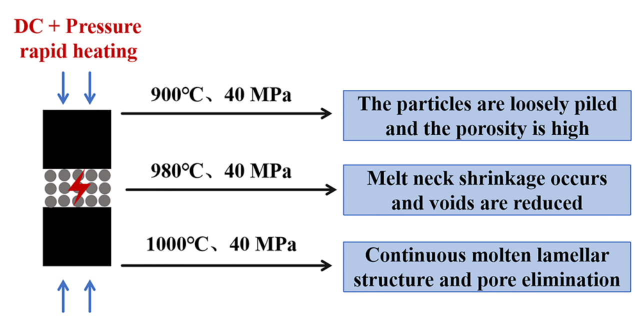

Figure 6. Schematic illustration of the FHPs process for lunar basalt consolidation.

The applied axial pressure (40 MPa) serves dual functions, suppressing abnormal grain growth during high-temperature processing and enhancing densification through concurrent plastic deformation and diffusion creep mechanisms. This yields a bulk material with 95% relative density while maintaining a fine-grained microstructure. (3) Electric field-activated interface engineering: Lowers grain boundary energy barriers, facilitating elemental (Si, Al, Ca) migration; the formation of a continuous silicate network

Figure 7. Macroscopic view of the large-format basalt sample (210 mm × 110 mm × 10 mm) fabricated via field-assisted hot-pressed under optimized conditions (960°C, 30 MPa and 20 min).

framework is promoted; the interfacial bonding in ferromagnesian mineral phases is strengthened. Figure 6 schematically illustrates this multi-physics processing approach for lunar basalt consolidation.

Building upon these findings, the research team successfully fabricated a large-format basalt sample (210 mm × 110 mm × 10 mm) under optimized processing conditions (960°C, 30 MPa and 20 min) in figure 7. This breakthrough not only confirms the tunability and reliability of FHPs processing parameters but also demonstrates significant engineering potential for scalable applications, laying a critical foundation for industrial-scale use of basalt in lunar construction elements. Notably, the energy consumption of the FHPs process was quantified at 2.8 ± 0.3 kWh kg^{-1} for the 960°C/30 MPa/20 min condition, significantly lower than for conventional hot-pressed (5–8 kWh kg^{-1}) and laser sintering (15–20 kWh kg^{-1}). This energy efficiency advantage is critical for ISRU, as lunar power generation is limited by solar panel capacity and storage constraints. This systematic investigation, spanning compositional analysis to processing innovation, marks a pivotal transition in lunar basalt research from fundamental science to practical engineering. As international lunar exploration advances, future work will focus on three key directions. The elucidating genetic links between lunar mantle composition and magmatic evolution, developing intelligent adaptive manufacturing technologies for extreme extraterrestrial environments, and engineering advanced functional materials (e.g., basalt fiber composites, porous ceramics). These developments will not only provide essential material solutions for lunar infrastructure but also establish a universal technological framework for ISRU on extraterrestrial bodies.

4. Conclusions

This study demonstrated the successful fabrication of lunar basalt bulk materials that achieved 95% relative density through the DC FHPs method in the range of 900°–1000°C.

(1) The optimal mechanical properties were achieved under the sintering parameters of 1000°C, 40 MPa, and 10 min. These properties included a high bulk density of 3.12 g cm^{-3}, a superior Vickers hardness of 853.7 HV, an enhanced fracture toughness (K_{IC} = 2.2 MPa $m^{1/2}$), and a significant bending strength of 85 MPa.
(2) The key novelty lies in demonstrated rapid densification within 10–20 min at 960°–1000°C, achieving 95% density-significantly faster than the conventional hot-pressed method (typically 1 h).
(3) This work clarified temperature-dependent mechanisms for densification and grain evolution. Findings provided key parameters for *in-situ* use of lunar basalt, aiding future fabrication of construction materials for lunar bases.

Acknowledgements

This work was funded by the Major Basic Research Project of the Natural Science Foundation of the Jiangsu Higher Education Institutions (Grant Nos. 23KJA430016), JMRH innovation platform for R&D of titanium matrix composites for aeroengines in Jiangsu Province (Grant Nos. JPTC2406) and Guangxi Key Laboratory of Advanced Structural Materials and Carbon Neutralization Open Fund Project (Grant Nos. GXAMCN25-5).

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